

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1510

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

Total Marks: 20

Question Count: 4

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

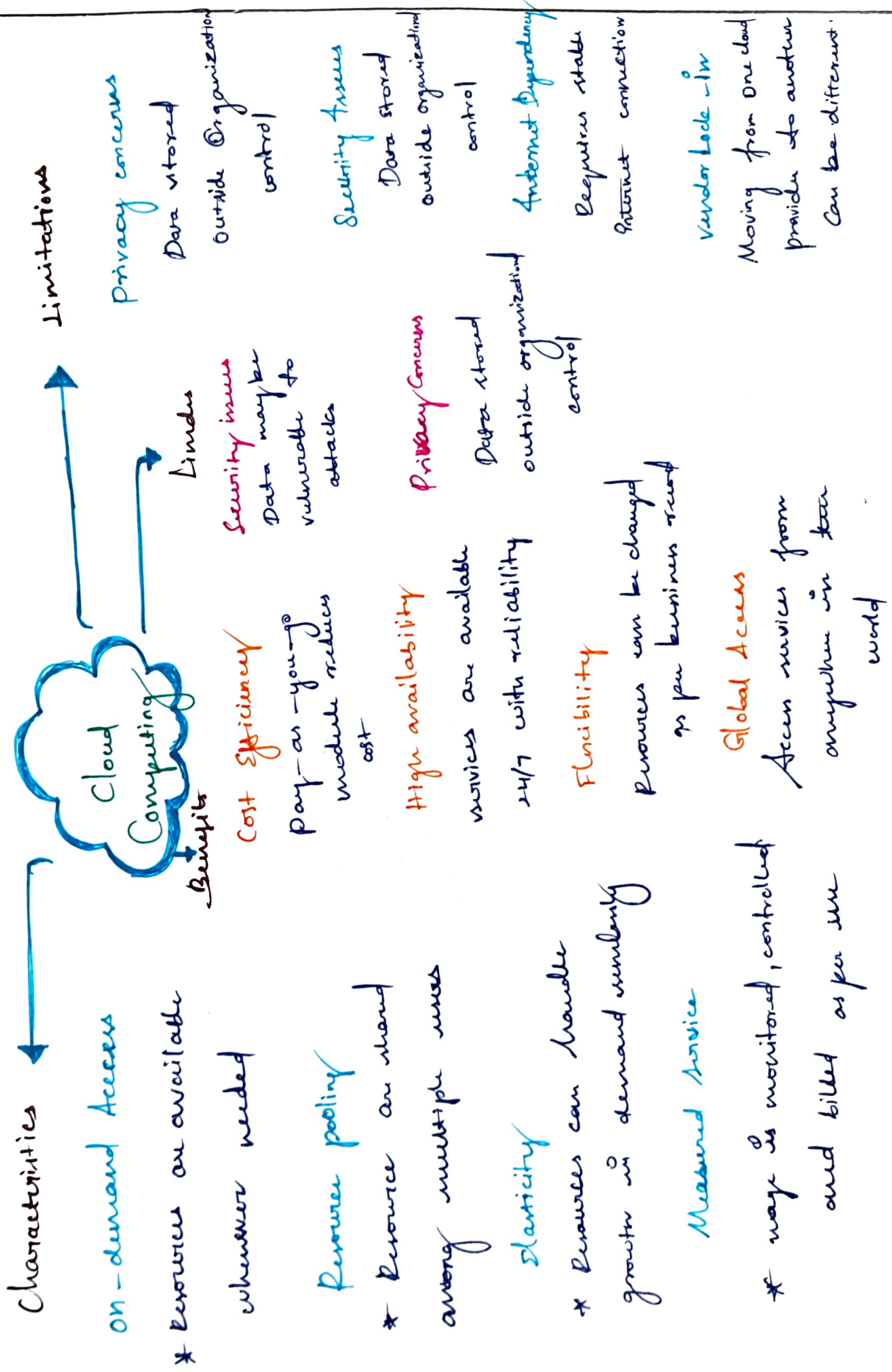
Code of Conduct

I Prabharathi P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prabharathi P

Concept Map 1 : Cloud Computing Fundamentals



Concept Map & Evolution of cloud computing

Hardware Evolution

- * Mainframes
- * Personal Computers
- * Servers
- * Virtualization Technology

↓
Leads to

Efficient use of hardware
better resources utilization
through virtualization

Internet & software

Evolution

- * Internet growth
- * Web application
- * APIs and web services

↓
enable

Easy access, connectivity
and development of
internet based services

Distributed System

- * Grid Computing
- * Peer-to-peer systems
- * Utility computing

↓
enable

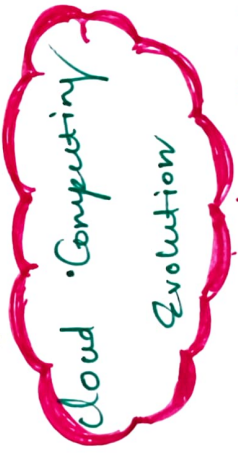
Resource sharing
scalability
Fault tolerance

Business needs

- * Cost Reduction
- * Agility
- * Flexibility
- * Global Reach

↓
drives

Demand for
on-demand services
and outsourcing
IT resources



↓
results in

Security Delivery Models

IaaS

(Infrastructure as a service)

PaaS

(Platform as a service)

SaaS

(Software as a service)

XaaS

(Everything as a service)

Concept Map 3: Service Models

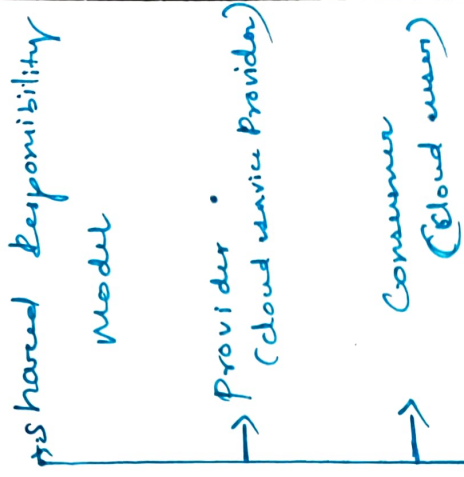
Cloud Service Models

IaaS
(Infrastructure as a service)
↓ provides ↓
Infrastructure Layer
(Servers, storage, Network, Virtual machines)
↓
Provider Responsibilities
Manage physical infrastructure, servers, storage, networking, virtualization.
↓
User Responsibilities

PaaS
(Platform as a service)
↓ provides ↓
Platform Layer
(OS, Runtime, Database, Development tools)
↓
Provider Responsibilities
Manage infrastructure, OS, runtime, middleware, database, tools
↓
User Responsibilities

SaaS
(Software as a service)
↓ provides ↓
Application layer
(Complete software applications)
↓
Provider Responsibilities
Manage everything (infrastructure, platform and application)
↓
User Responsibilities

XaaS
(Everything as a service)
↓ provides ↓
Any service delivered as a service over cloud
↓
Provider responsibilities
Delivers and manages the services
↓
User Responsibilities



but services an enabler of cloud

Visualization

Snatched by

Club services

Creating virtual machines (vms)

VMs run on hypervisor

Hyperminors provide abstraction of hardware.

Shuttle Busse Pooling and multi-tenancy

Support, Availability, Elasticity and Flexibility

Reduces cost and

Empirical Resource Utilization

→ Anabals

cloud computing

enables

↑
Together
Erasmus

Cloud Deployment and operations

~~Ph~~ On-demand Access

- * Clarity

-x measured twice

Stability

~~x~~ knowen
pooling

- * High Availability

APIs Implemented using

we APIs for Communication

```

graph LR
    Soap[Soap] --- Rest[Rest]

```

Connect . Provider and
Consumer via Internet

Enable Performanceability

and loss coupling

Facilitate On-demand

Yavica delivery